

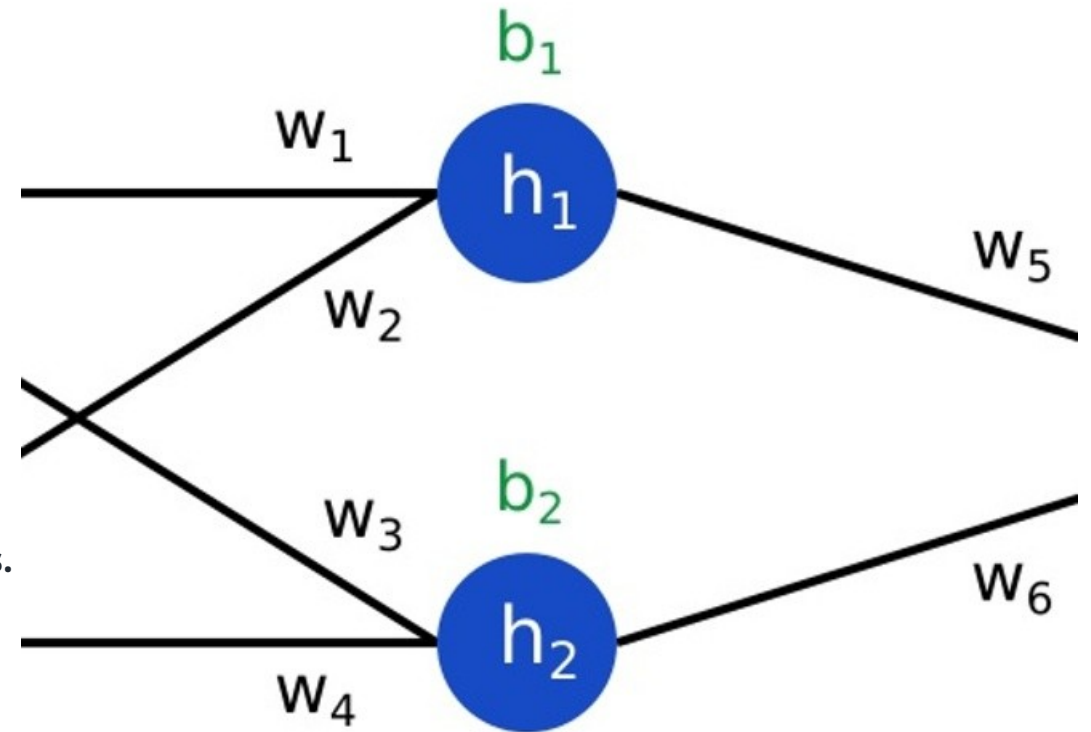
OpenCV Overview

Open-source computer vision library

KEY TAKEAWAY

OpenCV is a versatile, free library for computer vision across many languages.

Hidden Layer



From Inception to Today

Evolution of OpenCV

OpenCV has evolved from a research prototype to a widely adopted, cross-platform library.

Key Capabilities

What OpenCV can do



Image I/O

Read, write, and convert many formats.



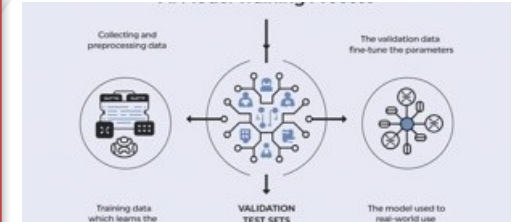
Filtering

Apply blur, edge detection, and morphology.



Feature Detection

Detect corners, blobs, and keypoints.



Machine Learning

Train classifiers and neural networks.

Library Structure

Organized functional modules

```
1 // ...
2
3 * Import java.util.ArrayList;
4 * Import java.util.Collections;
5 * Import java.util.Comparator;
6 * Import java.util.List;
7
8 * public class TwitterRepository {
9 *     private final List<TwitterUser> twitterUsers = new ArrayList<>();
10 *
11 *     public void add(TwitterUser twitterUser) {
12 *         twitterUsers.add(twitterUser);
13 *     }
14 *
15 *     public List<TwitterUser> getTwitterUsers() {
16 *         Collections.sort(twitterUsers, new Comparator<TwitterUser>() {
17 *             public int compare(TwitterUser u1, TwitterUser u2) {
18 *                 return u1.getTwitterMedia().compareTo(u2.getTwitterMedia());
19 *             }
20 *         });
21 *     }
22 *
23 *     return twitterUsers;
24 * }
```

Core

Fundamental data types and utilities.



Imgproc

Filters, transformations, and color conversion.



Video

Capture, encode, and decode video streams.



DNN

Run pre-trained deep neural networks.

Installation & Setup

Quick start for Python and C++

Reading and Displaying Images

First steps in code

Loading and showing images requires only three OpenCV functions.

Feature Detection & Matching

SIFT, SURF, ORB explained

Real-World Use Cases

From healthcare to autonomous cars



Healthcare

Segmentation of MRI and CT scans.



Autonomous Vehicles

Real-time lane and obstacle detection.



Retail

Shelf inventory analysis with cameras.



Augmented Reality

Overlay graphics on live video streams.